

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, suspension	Molecules Electroporated	cDNA's : supercoiled and linear.
Species Used	Human, HepG2, hepatoma; Monkey, COS-7, kidney.		

Before the Pulse

Cell Growth Medium	RPMI, 10% Fetal Calf Serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Per Bio-Rad protocol
		Pre-pulse Incubation	Per Bio-Rad protocol

Wash Solution Per Bio-Rad protocol (see Gene Pulser Instruction Manual)

The Pulse **Instruments Used** Gene Pulser® apparatus & Capacitance

Electroporation Temperature	0 °C (ice)		
Electroporation Medium*	Phosphate Buffered Saline	Cuvette Gap	0.2 cm
Cell Density	10 (7) cells / ml	Voltage	0.30 kV
Volume of Cells	0.4 ml	Field Strength	1.5 kV/cm
DNA Concentration	10 to 20 µg	Capacitor	960 µF
DNA Resuspension Buffer	Not given	Resistor	(Pulse Controller) Ω none
Volume of DNA	1 µl to 10 µl	Time Constant	Not given

After the Pulse
Outgrowth Medium RPMI /10% Fetal Calf Serum

Relevant Publications and/or Comments
Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	Not given
Selection Method or Assay Used	FACS
Electroporation Efficiency	100 transfectants/ µg DNA
Per Cent Survival	30 to 50%

Name of Submitter J.S. Maltor

Institution Address Tulane Medical Center
Pathology Dept.
1430 Tulane Avenue
New Orleans, LA 70112

Survey Number
161